

Package Management

Objectives

After completing this lesson, you should be able to:

- Describe Oracle Linux package management
- Use the `rpm` utility
- Describe the Oracle Linux yum server
- Explain and configure yum repositories
- Use the `yum` utility
- Describe the Unbreakable Linux Network (ULN)
- Describe how to register a system with ULN
- List the steps to switch from Red Hat Network (RHN) to ULN
- Explain replacement of SSL certificates for ULN
- Describe Software Collections and how to use them



Package Management: Introduction

- Oracle Linux uses Red Hat Package Manager (RPM).
- Oracle Linux also provides the `yum` utility, which:
 - Resolves RPM dependencies
 - Connects to repositories to download software
- The Oracle Linux yum server:
 - Offers a free way to install packages
 - Has free errata
- Unbreakable Linux Network (ULN):
 - Is a comprehensive resource for support subscribers
 - Offers access to software patches, updates, and fixes



The rpm Utility

- Query options:

```
# rpm -qa  
# rpm -qi package_name  
# rpm -ql package_name  
# rpm -qf filename  
# rpm -qc package_name
```

- Installing and updating packages:

```
# rpm -Uvh package_name
```

- Installing a new kernel:

```
# rpm -ivh kernel_package_name
```

- Removing packages:

```
# rpm -e package_name
```

Oracle Linux Yum Server

- Oracle offers free packages from Oracle Linux installation media.
- Errata are also available from the Oracle Linux yum server.
 - Subscribe to the Oracle errata mailing list from this site.
- Packages can be accessed at:
 - <https://yum.oracle.com/>
- Repository files are used to provide access to yum repositories needed for your system.
 - Previously, the main repository file was `public-yum-ol7.repo`.
- A modular `yum` repository configuration provides multiple, more targeted repository files.
 - These repository files replace `public-yum-ol7.repo`.
 - Existing `public-yum-ol7.repo` files can still be used, but will no longer be updated.

The `yum` Configuration

- `/etc/yum.conf`:
 - Is the primary configuration file
 - Holds global settings
- `/etc/yum.repos.d`:
 - Is the directory that contains `repo` files
- `repo` files define which repositories to use.
- Each `repo` file includes specifications for related repositories.
- The `baseurl` directive indicates the location of the main repository.
- The `enabled` directive (set to 1) designates the repository to use.

Modular `yum` Repository Configurations

- In the past, the `public-yum-ol7.repo` file contained a large number of repository definitions for many RPMs.
 - As new packages were added, more repository definitions were added to this single file.
- A modular `yum` repository definition approach is now implemented.
 - Smaller, more targeted repository files are provided.
 - This makes it easier to enable specific repositories and keep `yum` repository definitions current.
- New release RPMs are available in the `ol7_latest` repository.
 - These release RPMs let you configure and update repositories for specific RPMs you are interested in.
- The `public-yum-ol7.repo` file will no longer be updated with new repository definitions.

Systems With Modular `yum` Repository Configurations

- You can verify whether your system is configured with the modular `yum` approach by checking for installation of the `oraclelinux-release-el7` RPM:

```
# rpm -q oraclelinux-release-el7
# oraclelinux-release-el7 ...
```

- You also will no longer have an active `public-yum-ol7.repo` file.
- Install other release RPMs you desire and `yum` repository files will be configured accordingly.
- To see a current list of installed and available release RPMs:

```
# yum list *release-el7
```


Modular `yum` Repository Configurations: Legacy Systems

- For legacy Oracle Linux installations with a monolithic `public-yum-ol7.repo` file, initial new repository files are obtained by:
 - Running `yum update`, installing the `oraclelinux-release-el7` RPM, or installing one of the other release RPMs
 - These repository files have a `.disabled` extension initially.
- The script `/usr/bin/ol_yum_configure.sh` is installed and must be run as `root`, to complete the transition.
 - Currently enabled repositories are checked.
 - Needed release RPMs are installed, providing further required repositories.
 - New repository files will then have `.repo` extensions and be active.
 - The `public-yum-ol7.repo` file will no longer be used.
- If no action is taken, the existing `public-yum-ol7.repo` file will work, but will not be updated.

The `yum` Utility

- List all packages:

```
# yum list
```

- List all installed packages:

```
# yum list installed
```

- List packages available to be installed:

```
# yum list available
```

- Check for updates to installed packages:

```
# yum check-update
```

- Update, install, and remove packages:

```
# yum update package_name  
# yum install package_name  
# yum remove package_name
```

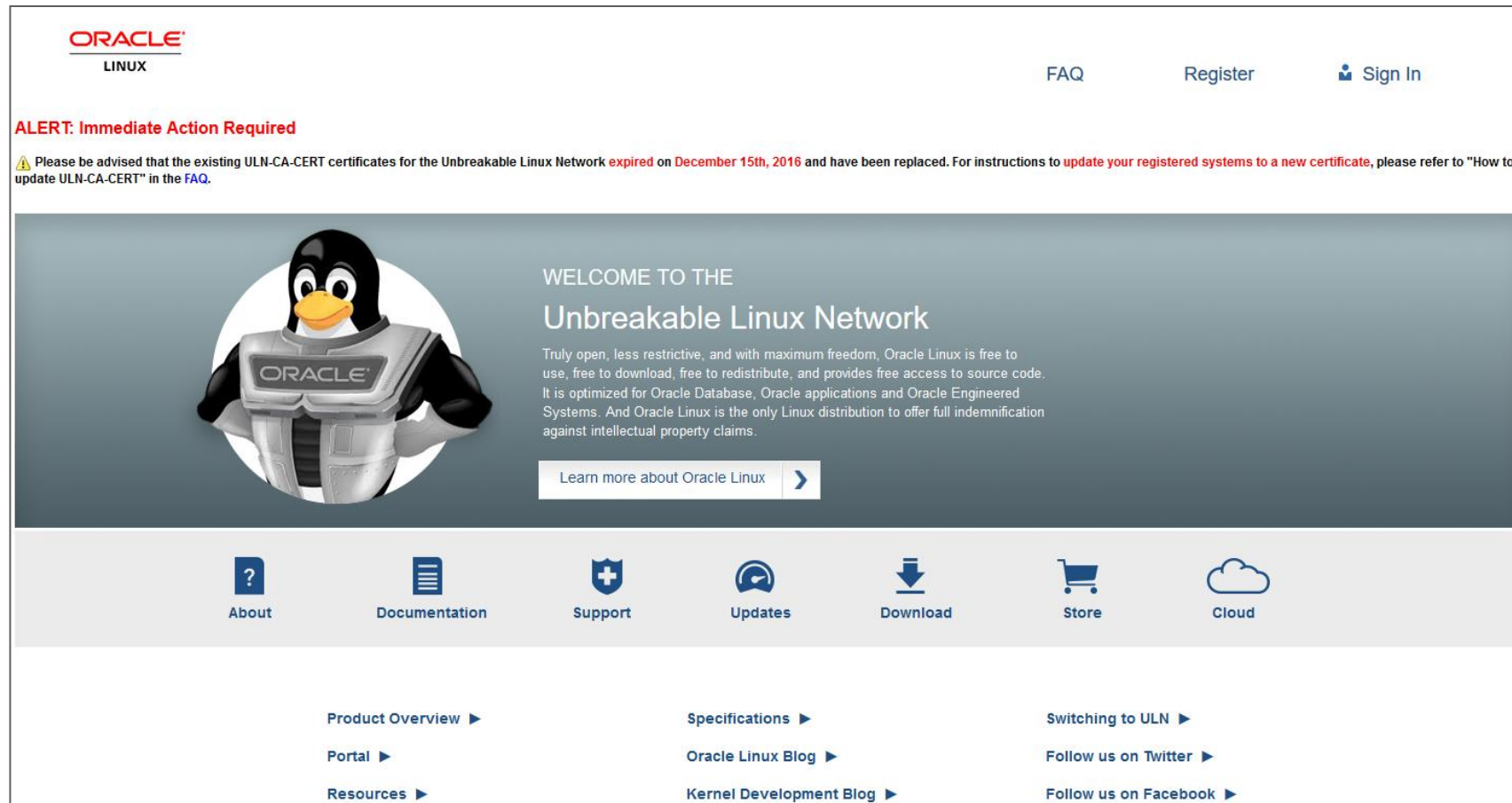
yum Groups

- `yum` groups are collections of software packages referred to by a single “group” name.
- `yum` supports the following group commands:

```
# yum group list
# yum group info groupname
# yum group install groupname
# yum group update groupname
# yum group remove groupname
```

Unbreakable Linux Network (ULN)

ULN is accessed at <https://linux.oracle.com/>.



ULN Channels

Register your systems and subscribe to a ULN channel.



Unbreakable Linux Network (ULN)

Welcome: [Search](#) [Profile](#) [About](#) [SSO Logout](#)

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Channels

Release Oracle Linux 7 Architecture x86_64

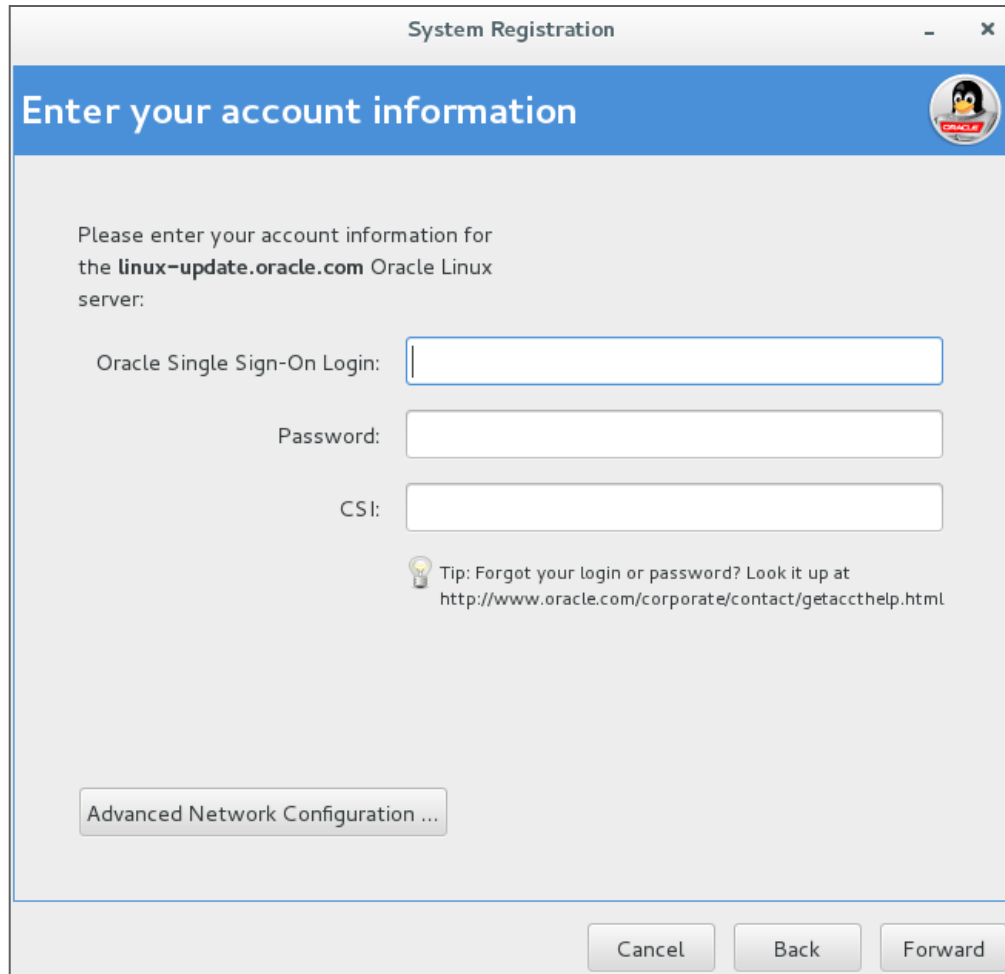
Name	Label	Description	Packages
Oracle Linux 7 Latest Optional Packages (x86_64)	ol7_x86_64_optional_latest	All optional packages released for Oracle Linux 7 (x86_64) including the latest errata packages. (x86_64)	5042
Oracle Linux 7 Latest (x86_64)	ol7_x86_64_latest	All packages released for Oracle Linux 7 (x86_64) including the latest errata packages. (x86_64)	5164
Oracle Linux 7 Archive (x86_64)	ol7_x86_64_latest_archive	Archived packages for Oracle Linux 7 (x86_64)	5220
Oracle Linux 7 Update 1 installation media copy (x86_64)	ol7_x86_64_u1_base	All packages released for Oracle Linux 7 Update 1 (x86_64). No errata included	4384
Oracle Linux 7 Update 1 Patch (x86_64)	ol7_x86_64_u1_patch	Updated packages published after release of Oracle Linux 7 Update 1 (x86_64)	531
Oracle Linux 7 GA installation media copy (x86_64)	ol7_x86_64_u0_base	All packages released for Oracle Linux 7 GA (x86_64). No errata included	4315
Oracle Linux 7 Update 2 installation media copy (x86_64)	ol7_x86_64_u2_base	All packages released for Oracle Linux 7 Update 2 (x86_64). No errata included	4634
Oracle Linux 7 Update 2 Patch (x86_64)	ol7_x86_64_u2_patch	Updated packages published after release of Oracle Linux 7 Update 2 (x86_64)	685
Oracle Linux 7 GA Patch (x86_64)	ol7_x86_64_u0_patch	Updated packages published after release of Oracle Linux 7 GA (x86_64)	583
Oracle Linux 7 Update 3 installation media copy (x86_64)	ol7_x86_64_u3_base	All packages released for Oracle Linux 7 Update 3 (x86_64). No errata included	4768
Oracle Linux 7 Update 3 Patch (x86_64)	ol7_x86_64_u3_patch	Updated packages published after release of Oracle Linux 7 Update 3 (x86_64)	546
Oracle Linux 7 Update 4 installation media copy (x86_64)	ol7_x86_64_u4_base	All packages released for Oracle Linux 7 Update 4 (x86_64). No errata included	4999
Oracle Linux 7 Update 4 Patch (x86_64)	ol7_x86_64_u4_patch	Updated packages published after release of Oracle Linux 7 Update 4 (x86_64)	703
Oracle Linux 7 Update 5 installation media copy (x86_64)	ol7_x86_64_u5_base	All packages released for Oracle Linux 7 Update 5 (x86_64). No errata included	5127
Oracle Linux 7 Update 5 Patch (x86_64)	ol7_x86_64_u5_patch	Updated packages published after release of Oracle Linux 7 Update 5 (x86_64)	659

Switching from RHN to ULN

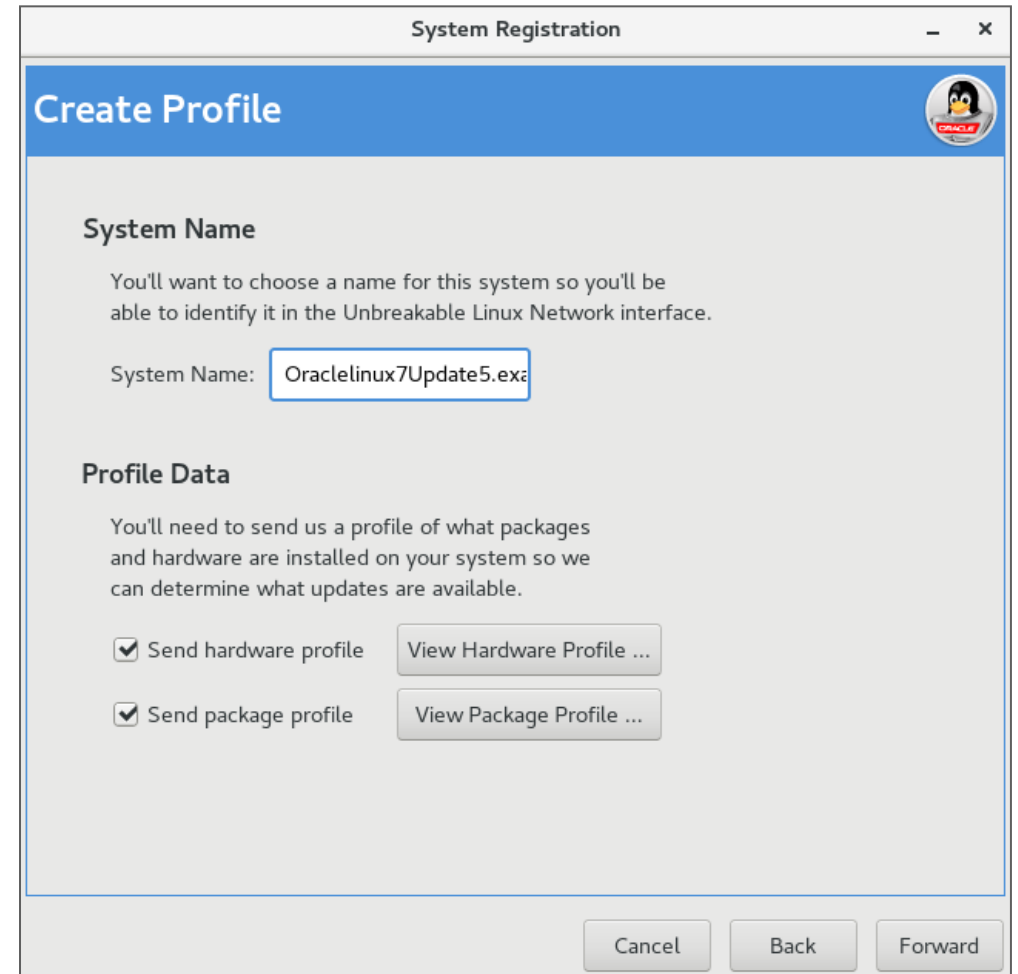
- RHEL uses the `rhn_register` program to register a system with Red Hat Network (RHN).
- Oracle has a `uln_register` program to switch to ULN if you are running RHEL.
- Do the following:
 1. Download the following files from <https://linux-update.oracle.com/rpms>:
 - `uln_register_ol7.tgz`
 - `uln_register-gnome_ol7.tgz`
 2. Extract the packages.
 3. Install the packages.
 4. Create a ULN account at <https://linux.oracle.com/register>.
 5. Register your system with ULN by running `uln_register`.

Registering Your System with ULN

These are the initial screens when you run the `uln_register` command from a terminal window in an X Window system.



The screenshot shows the 'System Registration' window with the title 'Enter your account information'. It features a blue header bar with the Oracle Linux logo. The main content area has a light gray background. The text 'Please enter your account information for the **linux-update.oracle.com** Oracle Linux server:' is displayed. Below this, there are three input fields: 'Oracle Single Sign-On Login:', 'Password:', and 'CSI:'. A tip icon (lightbulb) is present next to the text 'Tip: Forgot your login or password? Look it up at <http://www.oracle.com/corporate/contact/getaccthhelp.html>'. At the bottom left, there is a button labeled 'Advanced Network Configuration ...'. At the bottom right, there are three buttons: 'Cancel', 'Back', and 'Forward'.



The screenshot shows the 'System Registration' window with the title 'Create Profile'. It features a blue header bar with the Oracle Linux logo. The main content area has a light gray background. The text 'You'll want to choose a name for this system so you'll be able to identify it in the Unbreakable Linux Network interface.' is displayed. Below this, there is a label 'System Name:' followed by an input field containing the text 'Oraclelinux7Update5.exa'. Below the input field, there is a section titled 'Profile Data' with the text 'You'll need to send us a profile of what packages and hardware are installed on your system so we can determine what updates are available.' Below this, there are two checkboxes: 'Send hardware profile' and 'Send package profile', both of which are checked. To the right of each checkbox is a button labeled 'View Hardware Profile ...' and 'View Package Profile ...' respectively. At the bottom right, there are three buttons: 'Cancel', 'Back', and 'Forward'.

Replacement of SSL Certificates for ULN

- Oracle has replaced Symantec-branded certificates with Digicert-branded certificates across all of its infrastructure.
 - Change in certificates occurred on October 9, 2018.
- It requires that client certificates on all Oracle Linux systems directly subscribed to and receiving updates from ULN be updated.
- Connection to ULN will not be possible after October 9, 2018 until certificates are updated.
- For Oracle Linux 7, the following packages must be updated to these versions or later:
 - `rhn-client-tools-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-setup-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-check-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-setup-gnome-2.0.2-21.0.9.el7.noarch.rpm` (only required if a previous version is already installed)
 - If these packages were installed before October 9, 2018, no further action is required.

Replacement of SSL Certificates for ULN

- If the four packages were not updated as indicated before October 9, 2018, connection to ULN will not be possible until certificates are manually updated.
- In this case, the SSL certificate on Oracle Linux systems must first be replaced by following these steps:

```
# cp /usr/share/rhn/ULN-CA-CERT /usr/share/rhn/ULN-CA-CERT.old  
# wget https://linux-update.oracle.com/rpms/ULN-CA-CERT.sha2  
# cp ULN-CA-CERT.sha2 /usr/share/rhn/ULN-CA-CERT
```

Replacement of SSL Certificates for ULN

- After manually replacing the SSL certificate on your Oracle Linux systems, install the four packages as part of your patching cycle:
 - `rhn-client-tools-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-setup-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-check-2.0.2-21.0.9.el7.noarch.rpm`
 - `rhn-setup-gnome-2.0.2-21.0.9.el7.noarch.rpm` (only required if a previous version is already installed)
 - This presumes these four packages were not installed before October 9, 2018.
- Systems managed by Oracle Enterprise Manager or that are subscribed to a local Spacewalk instance are unaffected by this SSL certificate issue.

The Software Collection Library for Oracle Linux

- Set of software providing support mainly for developers
- Provides later versions of software than are available from a standard Oracle Linux installation (for example: GCC, Python, Perl, and `strace`)
 - Developers may need to work with more current features that are available.
- Software from a collection is installed and executed separately from versions installed with Oracle Linux.
- Collections are available for Oracle Linux 6 and Oracle Linux 7. Check for differences as necessary.
- Packages are available on the ULN and Oracle Linux yum server.
- Example collection: `devtoolset-7`
 - Set of tools for developers

Using Software Collections with Oracle Linux 7

- Ensure you are subscribed to the following ULN channels:
 - `ol7_x86_64_latest`, `ol7_x86_64_optional_latest` and `ol7_x86_64_SoftwareCollections`
- Or ensure the following Oracle Linux yum server repositories are enabled:
 - `ol7_latest`, `ol7_optional_latest` and `ol7_software_collections`
- To view available packages within the ULN software collections channel, run:

```
# yum --disablerepo="*" --enablerepo="*SoftwareCollections" list available
```

- To view available packages within the yum software collections repository, run:

```
# yum --disablerepo="*" --enablerepo="*software_collections" list available
```

- To see software collections packages that are already installed from the yum software collections repository, run:

```
# yum list installed|grep software_collections
```

Installing Software Collections and Running Commands

- The `scl-utils` package must be installed in order to run software from a collection, if not installed already:

```
# yum install scl-utils
```

- The `scl` utility from the `scl-utils` package is used to execute commands or utilities from a software collection with the `enable` action.
- Use `yum` to install a collection. For example, to install Developer Toolset version 7:

```
# yum install devtoolset-7
```

- Software collection packages are installed in the `/opt` path.
- `strace` is one of the utilities in the `devtoolset-7` collection. Use the `scl` utility to run the `devtoolset-7` software collection version of `strace`. Example:

```
# scl enable devtoolset-7 -- strace -e trace=/stat$ ls /boot/vmlinuz*
```

To view man pages for software collections, use the `scl` utility as in these examples:

```
# scl enable devtoolset-7 'man devtoolset-7'  
# scl enable devtoolset-7 'man strace'
```

Creating a Shell Environment and Viewing Optional Packages

- Commands or utilities associated with one or more collections can be made available in a shell environment without running the `scl` utility within the shell. Example:

```
# scl enable devtoolset-7 bash
```

- The path to the software collection version of `strace` is known within the shell:

```
# which strace  
/opt/rh/devtoolset-7/root/usr/bin/strace
```

- To enable two or more software collections within the same shell environment, list them on the command line. Example:

```
# scl enable devtoolset-7 python33 bash
```

- Collections can have optional packages that are not installed with the main package and its dependencies.
 - Use the `yum` command to see available optional packages. Example:

```
# yum list available devtoolset-7-\*
```

Quiz



Yum repositories can be both local and remote.

- a. True
- b. False



Quiz



Errata are available only by subscribing to ULN.

- a. True
- b. False



Quiz



Which of the following statements about the Software Collection Library for Oracle Linux are true? (Select all that apply.)

- a. The collections are primarily intended for developers.
- b. The collections provide later versions of software than are available from an Oracle Linux installation.
- c. Packages from the collections are installed in the same directory path as packages from an Oracle Linux installation.
- d. Packages are available from the Unbreakable Linux Network (ULN) and the Oracle Linux yum server.



Summary

In this lesson, you should have learned how to:

- Describe Oracle Linux package management
- Use the `rpm` utility
- Describe the Oracle Linux yum server
- Explain and configure yum repositories
- Use the `yum` utility
- Describe the Unbreakable Linux Network (ULN)
- Describe how to register a system with ULN
- List the steps to switch from Red Hat Network (RHN) to ULN
- Explain replacement of SSL certificates for ULN
- Describe Software Collections and how to use them



Practice 7: Overview

This practice covers the following topics:

- Using the `rpm` utility
- Accessing the Oracle Linux yum server
- Creating a local `yum` repository
- Using the `yum` utility
- Using Oracle Linux Software Collections
- Using the Unbreakable Linux Network (ULN)